MIL-V-58039E
26 March 1984
SUPERSEDING
MIL-V-58039D
2 November 1977

MILITARY SPECIFICATION

VALVES, GATE, RISING STEM,

DOUBLE ACTING, ALUMINUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 3-, 4-, and 6-inch double acting, rising stem, aluminum, gate valves (lightweight) for use with liquid petroleum supply systems.

1.2 Classification. The gate valves shall be the following sizes and types as specified (see 6.2):

Size - 3-inch.
  Type I - Round flanged ends.
  Type II - Threaded ends (internal)
Size - 4-inch.
  Type I - Round flanged ends.
Size - 6-inch.
  Type I - Round flanged ends.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225/9	- Aluminum Alloy Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finished, 7075.
QQ-A-601	- Aluminum Alloy Sand Castings.
QQ-B-626	- Brass, Leaded, and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-P-416	- Plating, Cadmium (Electrodeposited).
QQ-W-321	- Wire, Copper Alloy.

MILITARY

MIL-V-3	- Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-C-6183	- Cork and Rubber Composition Sheet; For Aromatic Fuel and Oil Resistant Gaskets.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-24396	- Packing Material, Braided Type TFE (Polytetrafluoroethylene).
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

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STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Colors.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 Sixteenth Street, NW, Washington, DC 22036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 135 - Seamless Brass Tube.
ASTM D 3951 - Standard Practice for Commercial Packaging,

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The gate valves shall be the contractor's standard commercial product, except as specified herein. The area of the flow passage, with the valve fully opened, shall be not less than that of a circle having a diameter equal to the nominal pipe size of the valve. The stem shall be a double acting, rising stem, connected to a wedge-shaped double disk and actuated by a manual handwheel to provide a positive valve action for fast closing. The double disk shall insure self-alinement when the valve is fully closed. The valve shall have a back seat to permit repacking of the stuffing box while the valve is in service and operating under its maximum rated pressure.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Materials for the gate valves shall be as specified herein (see 6.4).

3.3.1 Material deterioration and control. The gate valves shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Aluminum alloy.

3.3.2.1 Castings. The aluminum castings shall conform to QQ-A-601, alloy 356.0 or alloy 712.0.

3.3.2.2 Bars and rods. The aluminum bar or rod shall conform to QQ-A-225/9, temper T6.

3.3.3 Brass.

3.3.3.1 Rod. The brass rod shall conform to QQ-B-626, copper alloy number 360, temper half-hard.

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3.3.3.2 Forging. The brass forging shall conform to QQ-B-626, copper alloy number 377.

3.3.3.3 Tube. The brass tube shall conform to ASTM B 135, copper alloy UNS No. C23000, temper H80.

3.3.4 Casting, bronze. The bronze casting shall conform to QQ-C-390, copper alloy number 844, condition as cast.

3.3.5 Wire, phosphor-bronze. The wire shall be phosphor-bronze conforming to QQ-W-321, alloy number 510.

3.3.6 Packing, valve stem. The packing for the valve stem shall conform to MIL-P-24396, type A.

3.3.7 Gasket, cork and rubber composition sheet. The cork and rubber composition gasket shall conform to MIL-C-6183 type I or II, class 1, grade B or C.

3.3.8 Capscrews or bolts and nuts and lockwashers. The capscrews or bolts and nuts and lockwashers shall be fabricated of steel, cadmium plated in accordance with QQ-P-416, class 2.

3.4 Performance. The gate valve shall be lightweight to handle liquid petroleum products at working pressures of 125 pounds per square inch (psi). When tested as specified in 4.5.2.1.1, there shall be no distortion, visible porosity of the shell or leakage or seepage around the stem. When tested as specified in 4.5.2.1.2, there shall be no leakage or seepage through or around the disk or seat rings. When tested as specified in 4.5.2.1.3, there shall be no leakage or seepage through or around the packing. The disk shall not jam and internal parts shall not be damaged by excessive tightening of the handwheel as specified in 4.5.2.1.4.

3.5 Interchangeability. All valves furnished under the same contract shall be identical to the extent necessary to insure interchangeability of valve trim normally identified by the contractor as replaceable. To insure interchangeable installation of valves in fixed assemblies, the distance from the vertical centerline of the valve to the flange contact surface shall not vary by more than plus or minus 1/32-inch from the contractor's standard dimension.

3.6 Construction.

3.6.1 Body. The valve body shall be of cast aluminum alloy as specified in 3.3.2.1.

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3.6.1.1 Size 3-, 4-, and 6-inch, type I. The end flanges (inlet and outlet) of the valve body shall be integral with the valve body and shall be machine-faced with two V-shaped concentric grooves machined inside the bolt circle diameter. The flanges for the size 3-inch valve shall be 3/8-inch thick by 5-5/8-inch outside diameter with eight 7/16-inch diameter boltholes equally spaced to straddle the vertical and horizontal center lines on a 4-7/8-inch diameter bolt circle. The flanges for the size 4-inch valve shall be 3/8-inch thick by 6-3/4-inch outside diameter with eight 7/16-inch diameter boltholes equally spaced to straddle the vertical and horizontal center lines on a 5-7/8-inch diameter bolt circle. The flanges for the size 6-inch valve shall be 3/8-inch thick by 9-inch outside diameter with twelve 7/16-inch diameter boltholes equally spaced to straddle the vertical and horizontal center lines on an 8-1/8-inch diameter bolt circle. All dimensions shall have tolerances of plus or minus 1/64-inch.

3.6.1.2 Size 3-inch, type II. The inlet and outlet connections of the valve body shall be American Standard taper pipe thread (internal) in accordance with FED-STD-H28.

3.6.2 Bonnet. The bonnet shall be cast from aluminum alloy as specified in 3.3.2.1. The bonnet shall be an integral flange and shall be bolted to the mating flange of the valve body using gaskets, capscrews, or bolts and nuts and lockwashers as specified in 3.3.7 and 3.3.8. The bonnet shall be constructed to contain the stuffing box assembly (see 3.6.3) and shall allow adequate clearance for the disks when the gate valve is in the fully opened position.

3.6.3 Stuffing box. The stuffing box shall be self-adjusting (spring loaded) and shall consist of the following components.

3.6.3.1 Nut, packing. The packing nut shall be a material as specified in 3.3.3.1, 3.3.3.2, or 3.3.4. The internal thread shall be a standard thread size.

3.6.3.2 Gland, packing. The packing gland shall be material as specified in 3.3.3.1, 3.3.3.2 or 3.3.4. The gland may also be made from acetyl homopolymer (Dupont delrin or equal) or from acetyl copolymer (Celanese celcon or equal).

3.6.3.3 Spring, gland. The gland spring shall be phosphor bronze wire as specified in 3.3.5.

3.6.4 Packing. The packing shall be material specified in 3.3.6.

3.6.5 Stem, rising. The rising stem shall be material as specified in 3.3.3.1, 3.3.3.2, 3.3.3.3 or, for size 6-inch, of corrosion-resisting steel, AISI types 303, 304, 410, or 420, and shall be one or two piece construction

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having a double acting feature for fast closing. The stem shall include a back seat for repacking the stuffing box when the valve is fully opened and in service. The stem shall insure a positive grip on the keyed double-disk.

3.6.6 Disk, double. The double-disk shall be copper casting conforming to QQ-C-390, alloy 922, or material as specified in 3.3.3.2 or 3.3.4 or, for size 6-inch, of corrosion-resisting steel, AISI types 303, 304, 410, 416 or 420. The disk shall be the double-disk type with a self-alining feature, to insure a positive leak-proof closure. The disks shall be loosely keyed together to facilitate removal.

3.6.7 Ring, seat. The seat ring shall be material specified in 3.3.3.3 or, for size 6-inch, of corrosion-resisting steel as specified in 3.6.6. The bearing surfaces of the seat rings and double disks shall be lap finish. The method of seating of the seat rings shall be in accordance with the contractor's standard practice.

3.6.8 Handwheel. The handwheel shall be material specified in 3.3.2.1 and shall be the spoked type, with attachments to permit quick removal of the handwheel. The handwheels shall be marked "OPEN, CLOSE" with arrows to show direction of turning for opening and closing the valve.

3.7 Finish.

3.7.1 Anodic coating. Each aluminum component shall be anodized to a thickness of not less than 0.0007 inch before assembly and before testing as specified in 4.5.2. The coatings shall conform to MIL-A-8625, type II, dyed forest green in accordance with FED-STD-595, color No. 34079.

3.8 Marking for identification. Marking for identification shall be in accordance with MIL-STD-130, to include contractor's identification or trademark, size, rated pressure, fluid designation and the letters US.

3.9 Workmanship. Valves shall be free from imperfections and defects which may be injurious to the performance of the valve. There shall be no defective connections or loose, damaged or missing parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article valve shall be examined in accordance with 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article valve shall be tested as specified in 4.5.2.1.1, 4.5.2.1.2, 4.5.2.1.3 and 4.5.2.1.4. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level III.

4.4.1.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all gate valves of the same size and type offered for inspection at one time.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests. Each valve shall be tested as specified in 4.5.2.1.1, 4.5.2.1.2 and 4.5.2.1.3. Failure of any test shall be cause for rejection.

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4.5 Inspection procedure.

4.5.1 Examination. The gate valves shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Dimensions not as specified.
106. Components not anodized before assembly.
107. Finish not as specified.
108. Screws, nuts, and washers missing, broken, stripped, fractured, or loose.
109. Component not accessible for servicing.
110. Defective connections.
111. Loose, damaged, or missing parts.
112. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, illegible, or not affixed as specified.
202. Color of finish not as specified.

4.5.2 Tests.

4.5.2.1 Hydrostatic.

4.5.2.1.1 Valve body and back seat. The valve, assembled for actual operation, shall be subjected for not less than 1 minute to the following applicable hydrostatic pressure:

a. 3- and 4-inch valves - 300 psi.

b. 6-inch valve - 150 psi.

The valve shall then be closed and fully opened to seal the back seat. The packing shall then be removed and the valve re-pressurized as specified. Nonconformance to 3.4 shall constitute failure of this test.

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4.5.2.1.2 Valve disk and seat ring. The valve, with the disk closed, shall be subjected to the following applicable hydrostatic pressure on one end, while the disk and seat ring are examined through the opposite open end:

a. 3- and 4-inch valve - 125 psi.

b. 6-inch valve - 75 psi.

After initial pressure has been applied for a period of not less than 1 minute, the pressure shall be relieved and the valve opened and re-closed. The valve shall then be repressurized, as specified, on the opposite side while the disk and seat ring are examined through the opposing open end. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.1.3 Packing. The valve, assembled for actual operation with the disk open, shall be subjected to the following hydrostatic pressures for not less than 1 minute:

a. 3-inch and 4-inch valve - 125 psi.

b. 6-inch valve - 75 psi.

The stem shall then be rotated by the handwheel to remove contact from the back seat and the packing compressed by tightening of the gland so that the stuffing box or condensation chamber, if any, will be subjected to the hydrostatic pressure. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.1.4 Damage. The valve shall be closed by applying a force of not more than 2.5 foot pounds of torque to the handwheel. The torque pressure shall then be released and an opposing force of not more than 2.5 foot pounds of torque shall be applied to the handwheel to open the valve. The valve shall then be dismantled and examined. Nonconformance to 3.4 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with requirements of MIL-V-3, MIL-STD-129, and ASTM D 3951.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The valves shall be preserved in accordance with the level A preservation and packaging requirements of MIL-V-3.

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5.1.2 Commercial. The valves shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The valves shall be packed in accordance with the level A requirements of MIL-V-3, except fiberboard containers are not acceptable.

5.2.2 Level B. The valves shall be packed in accordance with the level B, requirements of MIL-V-3, except fiberboard containers shall be weather resistant.

5.2.3 Commercial. The valves shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951; in addition, weight and cube of the pack shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The valves covered by this specification are intended primarily for use with petroleum handling systems transporting liquid petroleum products.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size and type required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2).

d. Lot size required when other than as specified (see 4.4.1.1).

e. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required the item shall be examined and should be a first produced valve. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the documents first article.

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6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodians:
Army - ME
Navy - MC

Review activities:
Navy - YD
DLA - CS

User activities:
Army - CE
Navy - CG

Preparing activity:
Army - ME

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-58039E	Valves, Gate, Rising Stem, Double Acting, Aluminum

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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